

Homework 4

Merit and Excellence problem solving · about 60 minutes

Each question graded N0–E8

Answers · tutor's working, not an official mark scheme

Show all working, and finish every proof with a sentence. Each part shows the highest grade it can give: [A], [M] or [E].

Quick review

- 1 Find the discriminant of $x^2 - 4x + 4 = 0$. How many roots are there? [A]

$16 - 16 = 0$: one repeated root ($x = 2$)

- 2 Solve $\log_2 x = 5$. [A]

$x = 2^5 = 32$

- 3 Factorise $6x^2 + x - 2$. [A]

$(3x + 2)(2x - 1)$

- 4 Write a quadratic with roots $\frac{1}{2}$ and 3, with whole-number coefficients. [A]

$(2x - 1)(x - 3) = 0$, $2x^2 - 7x + 3 = 0$

QUESTION ONE

- (a) Show that the sum of four consecutive whole numbers is always even. [A]

$n + (n + 1) + (n + 2) + (n + 3) = 4n + 6 = 2(2n + 3)$, so it is always even

- (b) Prove that the product of an odd number and an even number is always even. [A]

$(2m + 1)(2n) = 2n(2m + 1)$, which is 2 times a whole number, so it is even

- (c) Prove that $(n + 3)^2 - (n - 3)^2$ is always a multiple of 12. [M]

$(n^2 + 6n + 9) - (n^2 - 6n + 9) = 12n$, so it is always a multiple of 12

- (d) Sam was asked to prove that the square of any odd number is one more than a multiple of 8. Sam's answer has 2 errors. Find each one, fix it, and say why it loses marks. [M]

Sam's answer

$3^2 = 9 = 8 + 1$ and $5^2 = 25 = 24 + 1$, so it is true.
Also $(2n + 1)^2 = 4n^2 + 1$.

1. two examples are not a proof: use $2n + 1$ for every odd number (L24)

2. $(2n + 1)^2 = 4n^2 + 4n + 1 = 4n(n + 1) + 1$; $n(n + 1)$ is even, so $8k + 1$ (L22)

- (e) On a calendar (7 days a row), a 3-by-3 square has top-left number n . Prove that the sum of all nine numbers is always 9 times the middle number. [E]

Rows: $n, n+1, n+2; n+7, n+8, n+9; n+14, n+15, n+16$

Sum = $9n + 72 = 9(n+8)$, and the middle number is $n+8$,

so the sum is always 9 times the middle number.

u: (a), (b) shown. r: (c) proved; (d) both errors fixed with reasons; (e) all nine numbers in terms of n .

t: (e) $9(n+8)$ with the conclusion. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION TWO

- (a) Find the points where $y = x^2 - 4$ meets $y = 3x$. [A]

$x^2 - 3x - 4 = 0, (x-4)(x+1) = 0$: points (4, 12) and (-1, -3)

- (b) Show that the line $y = 2x - 5$ never meets the curve $y = x^2 + 1$. [M]

$x^2 + 1 = 2x - 5, x^2 - 2x + 6 = 0, b^2 - 4ac = 4 - 24 = -20 < 0$

No real solutions, so the line never meets the curve.

- (c) Find the values of m for which $y = mx + 1$ is a tangent to $y = x^2 + 2x + 5$. [M]

$x^2 + (2-m)x + 4 = 0$; tangent: $(2-m)^2 - 16 = 0$

$2-m = \pm 4$: $m = -2$ or $m = 6$

- (d) Find every point where $x^2 + y^2 = 20$ meets $y = x^2$. [E]

$x^2 = y$, so $y + y^2 = 20, (y+5)(y-4) = 0$

$y = -5$ gives $x^2 = -5$: no real point. $y = 4$ gives $x = \pm 2$.

The points are (2, 4) and (-2, 4).

u: (a) both points. r: (b) shown with the discriminant; (c) both values.

t: (d) both points with $y = -5$ rejected for a reason. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

QUESTION THREE

- (a) Solve $\sqrt{2x+1} = 5$. [A]

$2x+1 = 25, x = 12$

- (b) Solve $\sqrt{x+12} = x$. [M]

$x^2 - x - 12 = 0, (x-4)(x+3) = 0$

Check: $x = -3$ gives $\sqrt{9} = 3 \neq -3$, rejected. $x = 4$

A rectangular pen is built against a wall, so 30 m of fencing makes only three sides: two of width x m and one of length $(30 - 2x)$ m.

- (c) The pen has area 100 m^2 . Find its possible dimensions. [M]

$$x(30 - 2x) = 100, \quad x^2 - 15x + 50 = 0, \quad (x - 5)(x - 10) = 0$$

5 m by 20 m, or 10 m by 10 m

- (d) Show that the pen cannot have area 120 m^2 . [M]

$$x^2 - 15x + 60 = 0: \quad b^2 - 4ac = 225 - 240 = -15 < 0, \text{ so no real width}$$

- (e) Use the discriminant to find the largest area the pen can enclose, and its dimensions. [E]

$$x(30 - 2x) = A, \quad 2x^2 - 30x + A = 0; \quad \text{real } x \text{ needs } 900 - 8A \geq 0$$

$A \leq 112.5$: the largest area is 112.5 m^2 , when $x = 7.5 \text{ m}$ and the length is 15 m .

u: (a) correct; (b) the quadratic formed. r: (b) $x = 4$ with -3 rejected; (c) both shapes; (d) shown.

t: (e) 112.5 m^2 with its dimensions and reasoning. N1 partial; N2 1u; A3 2u; A4 3u; M5 1r; M6 2r; E7 t, minor error; E8 t.

Totals out of 24. Cut scores move a little each year: Achievement from about 7–8, Merit 13–14, Excellence 19–20.